

Pharmacy GPT

AI-Powered Medicine Recommendation System

Project Documentation

Abstract

Pharmacy GPT is a web-based AI-powered medicine recommendation system developed using Python and Streamlit. The system accepts patient age, gender, symptoms, and medical notes, searches a disease dataset, and provides medicine recommendations focused on the Egyptian pharmaceutical market. When an unseen case is encountered, Ollama (Llama 3.2:3B) generates a structured medical record that is appended to the dataset, enabling dynamic knowledge-base expansion.

Introduction

The project aims to assist users by providing preliminary medicine recommendations, precautions, home-care advice, and follow-up guidance through an AI chatbot. It is an educational prototype and not a replacement for professional medical advice.

Objectives

- Analyze patient symptoms.
- Predict possible diseases.
- Recommend suitable OTC medicines.
- Show medicine availability and pricing in Egypt.
- Provide dosage guidance through the chatbot.
- Expand the dataset dynamically using Ollama.

System Workflow

1. User enters age, gender, symptoms and notes.
2. Dataset is searched.
3. If a match exists, the stored medical insight is used.
4. Otherwise Ollama generates a new structured record.
5. The new record is stored in the dataset.
6. AI generates recommendations and answers follow-up questions.

Technologies

Python, Streamlit, Pandas, LangChain, Ollama API (Llama 3.2:3B), JSON, CSV Dataset.

Dataset

Disease and Symptoms Dataset 2023 (Mendeley Data) is used as the project's medical knowledge source.

Architecture

Frontend: Streamlit

Backend: Python

Knowledge Base: Disease Dataset (CSV)

LLM: Ollama (Llama 3.2:3B)

Storage: Dynamic CSV updates

Future Work

Integrate authenticated user accounts, real-time pharmacy APIs, electronic prescriptions, multilingual support, and retraining pipelines.

Conclusion

Pharmacy GPT combines dataset-driven recommendations with an LLM to provide a smart educational pharmacy assistant capable of expanding its knowledge base over time.

Reference

Disease and Symptoms Dataset 2023, Mendeley Data:

<https://data.mendeley.com/datasets/2cxccsxydc/1>